Hybrid AI-Based Detection of Fileless Malware in IoMT Healthcare Systems

PRELIMINARY PROGRESS REVIEW (PPR)

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BSc (Hons) in Information Technology – Cybersecurity Specialization

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1. Introduction

1.1 Purpose of the Document

The following is a Preliminary Progress Report (PPR) that aims at formally documenting the present status, progress, and future plans of the hybrid artificial intelligence-based detection of fileless malware in the internet of medical things (IoMT) healthcare systems research project (Project Code: R26-CS-006). It shows that significant amounts of research and development work have been conducted according to PP1 guidelines and indicates how the rest will be completed.

The intended audience for this document includes the project supervisors, academic reviewers, and department panel members responsible for evaluating the preliminary progress of the research project.

1.2 Definitions, Acronyms, and Abbreviations

Acronym / Term	Definition
IoMT	Internet of Medical Things – networked medical devices in healthcare
AI	Artificial Intelligence
ML	Machine Learning
DICOM	Digital Imaging and Communications in Medicine – medical imaging protocol
PACS	Picture Archiving and Communication System
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations – XAI technique
LIME	Local Interpretable Model-agnostic Explanations – XAI technique
IDS	Intrusion Detection System
LSTM	Long Short-Term Memory – a type of recurrent neural network
XGBoost	Extreme Gradient Boosting – ensemble ML algorithm
APT	Advanced Persistent Threat
TTPs	Tactics, Techniques, and Procedures (MITRE ATT&CK framework)
MITRE ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge framework
PPR	Preliminary Progress Review
PP1	Progress Point 1 – first formal project milestone
RF	Random Forest – ensemble ML algorithm
SVM	Support Vector Machine

1.3 Overview

Organization of the PPR Document

The PPR document will be structured as follows:

- Section 2 provides an extensive literature review of current research on the security of IoMT technology, fileless malware, and AI-based detection.
- Section 3 discusses the research problem.
- Section 4 specifies the research questions.
- Section 5 articulates the research objectives.
- Section 6 discusses research methodology.
- Section 7 highlights the sources of data and analysis.
- Section 8 lists the anticipated benefits.
- Section 9 clarifies the research scope and deliverables.
- Section 10 outlines the research limitations.
- Section 11 lists the project schedule.
- References and Appendices are located at the end of the document.

2. Literature Review

2.1 IoMT Security Landscape

Internet of Medical Things (IoMT) is defined as the entire environment consisting of interconnected medical equipment, healthcare facilities, and telecommunications systems utilized in the present-day healthcare. Some examples of IoMT equipment include imaging equipment, infusion pumps, cardiac monitoring devices, wearable devices, and even hospital network infrastructure. While the connectivity of IoMT devices has led to an improvement in clinical efficiency and patient outcomes, it has also resulted in cybersecurity risks.

Coventry and Branley (2018) found that health facilities remain vulnerable to cyber attacks because of old equipment, poor patch management cycles, and poor network segregation. It is worth noting that the healthcare industry remains one of the most attacked sectors when it comes to cyber attacks. Breaches of patients' information and ransomware attacks lead to interruptions in the operation of health facilities and put the lives of patients at risk. Moreover, the integration of IT and OT technology in hospitals makes the situation even more complicated for security teams.

Fileless malware is another type of malware that poses an even greater threat in the IoT environment. Unlike the usual malicious software, which creates files on the hard disk of the infected computer, fileless malware works in memory, using legitimate system utilities, including PowerShell, Windows Management Instrumentation (WMI), and script interpreters, to run malicious programs in RAM. Fileless malware does not create any files that might trigger alerts from antivirus programs..

2.2 Fileless Malware Detection

Fileless malware is arguably one of the most advanced types of malware in today's cyber world. Based on the data presented by the Ponemon Institute (2020), the likelihood that fileless malware will be successful is about ten times higher than that for file-based malware because of its in-memory operation and reliance on legitimate processes. Fileless malware poses an even greater threat in the health care environment since most workstations run on Windows.

Current methodologies for fileless malware detection techniques mainly include three types: behavioral methods, memory forensics methods, and machine learning techniques. Behavioral methods include detecting malicious behaviors such as the creation of suspicious processes, network connections, and changes to the registry or scripts. The memory forensics methods entail analyzing the contents of volatile memory and identifying anomalies such as malicious code injection or suspicious process memory structures.

It is shown that using combined AI algorithms to detect fileless malware yields high efficiency. In the studies conducted where the combination of a recurrent neural network such as the LSTM model and the gradient boosting classifier was used, high results were obtained when processing the telemetry data of endpoints. This combination algorithm, with its advantages from the point of view of both classification based on features and sequencing, can be applied for multi-stage attacks via PowerShell

2.3 DICOM Medical Imaging Network Security

DICOM (Digital Imaging and Communication in Medicine), which is an internationally accepted protocol for handling medical imaging information, is employed universally in hospitals to transmit, store, and display medical images like those from computed tomography (CT) scans, magnetic resonance imaging (MRI) scans, and X-rays. The DICOM network usually consists of medical imaging equipment, Picture Archiving and Communication Systems servers, radiologists' workstations, and clinical review stations.

Despite the essential nature of DICOM in clinical decision-making, there has been scant research into the security of the DICOM environment, particularly from the perspective of cybersecurity. Studies like those of Mukherjee et al. (2020), among others, show that DICOM systems do not implement authentication, encryption, or access controls, making them vulnerable to attacks involving data exfiltration, image manipulation, and lateral movement. An attacker compromising the DICOM network can alter medical diagnostic images, extract patient information, or even utilize DICOM equipment to move around the hospital's network.

There are many machine learning based methods that have proven to be effective when it comes to detecting anomalies in network flows. The following features like packet flow rate, behavior of protocol being used, frequency of communication between source and destination nodes, and amount of data transferred become useful in this context.

2.4 Predictive Threat Forecasting in Healthcare Networks

The forecasting of attacks within IoMT frameworks involves predicting both the probability and type of attacks based on the behavioral patterns being experienced. While reactive detection solutions work to prevent attacks, threat forecasting helps create a proactive defense strategy because it detects signs of attack before the IoMT devices become compromised. According to Moustafa et al., behavioral traffic analysis employing ensembles could accurately predict the abnormal behavior of IoMTs.

CICIoMT2024 is an important database that has been widely used for analyzing IoT threats in healthcare settings. The use of the database in studies has included the analysis of Random Forest, XGBoost, and deep learning techniques in detecting IoMT anomalies, where some have achieved accuracy rates above 98%.

Time series prediction using LSTM learns beyond classification by capturing temporal behavior sequences, predicting future patterns of threats. This model fits well into the Internet of Medical Things, wherein the threat campaigns evolve over an extended period using multi-stage processes.

2.5 Lateral Movement Detection and Cross-Layer Correlation

The lateral movement process is an integral part of the cyber attack chain that involves the attacker's movement within the network infrastructure from the first point of compromise to more valuable endpoints. In the case of a healthcare system, the lateral movement will move laterally from one vulnerable IoMT endpoint to other internal networks and ultimately to clinical workstations or PACS.

The classic security tools such as the use of SIEMs (Security Information and Event Management) have tried to correlate events within different network levels; however, the drawback here is that they create numerous false positives and do not possess the intelligence that would be necessary to detect complex and multi-level attacks. The work done by Milajerdi et al. (2019) suggested the use of a graph-based provenance approach for attack path analysis within different system elements.

XAI has become highly relevant as a way of explaining AI-based decisions for human analysis. Methods like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) offer explanations at a feature level to help analysts understand the reasons behind labeling something as suspicious. The importance of XAI becomes clear when considering high-risk areas such as healthcare security.

2.6 Identification and Significance of the Research Problem

From the literature, it can be clearly observed that there is a serious and overlooked gap in the current body of research on IoMT security: The current state-of-the-art cybersecurity systems lack a cross-layer approach to AI detection that incorporates DICOM network anomaly detection, endpoint-level fileless malware detection, threat prediction, lateral movement tracking, and explainability.

The current security solutions tackle the issue in a piece-meal manner by considering each part separately. The network security solutions monitor network traffic but have no capabilities for endpoint monitoring. The endpoint security solutions lack the capability to detect any abnormalities with respect to the DICOM protocol as well as predict any threats for their further propagation.

This study aims to bridge this gap through the development and implementation of the Hybrid AI-based Detection of Fileless Malware in IoMT Healthcare Systems, which consists of four distinct modules that function both independently and in conjunction with one another.

3. Research Questions

The following research questions guide this investigation:

1. How can machine learning effectively detect anomalous behavior in DICOM medical imaging network traffic within IoMT healthcare environments?
2. What hybrid AI model architecture best combines behavioral classification and sequential pattern recognition for detecting PowerShell-based fileless malware from endpoint telemetry data?
3. Can behavioral traffic pattern analysis and predictive ML models reliably forecast IoMT threat probabilities with sufficient lead time for proactive response?
4. How can cross-layer security event correlation, combined with graph-based lateral movement analysis, identify multi-stage fileless attack paths traversing IoMT healthcare networks?
5. What XAI techniques can effectively make AI-driven lateral movement detection decisions interpretable and actionable for healthcare security analysts?

4. Research Objectives

4.1 Main Objective

The objective of this research project is to create and implement the Hybrid AI-Based Detection Framework for Fileless Malware in IoMT Healthcare Networks, which will include four interlinked modules providing multilayered malware detection, behavioral prediction, lateral movement correlation, and investigative support using explainable AI technologies for healthcare cybersecurity purposes.

4.2 Specific Objectives

- Develop a machine learning-based anomaly detection system for DICOM medical imaging network traffic, capable of identifying abnormal communication patterns indicative of data exfiltration, tampering, or lateral movement.
- Design and implement a hybrid AI model combining XGBoost and LSTM to detect PowerShell-based fileless malware from Sysmon endpoint telemetry, classifying malicious behavior and generating confidence-scored alerts.
- Build a predictive IoMT threat forecasting engine using Random Forest and LSTM-based time-series modeling to estimate threat risk probabilities and generate proactive security alerts with dashboard visualization.
- Implement a cross-layer lateral movement correlation detection engine that aggregates alerts from all detection modules, constructs attack progression graphs, and identifies multi-stage attack paths across IoMT infrastructure.
- Develop an XAI dashboard integrating SHAP and LIME explanations that presents detection results, lateral movement graphs, risk levels, and feature-level explanations in an accessible interface for security analysts.
- Evaluate all modules using quantitative metrics including accuracy, precision, recall, F1-score, and false positive rate, ensuring forensic soundness and suitability for healthcare security environments.

5. Research Methodology

The approach taken here is experimental and design-based. A modular design of the system allows independent development and evaluation of the detection elements prior to combining them into the comprehensive system. Tests for effectiveness of detection and integrity of the models are performed through datasets and simulations of IoMT attacks.

5.1 Component 1: DICOM Medical Imaging Network Anomaly Detection

Component 1 identifies any anomalies in the DICOM medical imaging communications system by detecting any malicious communication trends amongst the various medical devices, PACS servers, and clinical workstations.

This is done by first gathering IoMT network flow traffic data from the CICIoMT2024 dataset that consists of both normal and attack traffic in a healthcare environment. Features such as the type of protocol used, packet rates, flow durations, source-destination addresses, TCP/UDP flags, and data volume transferred will be collected and cleaned up for modeling purposes.

Classifiers such as Random Forest, XGBoost, and Isolation Forest, which utilize machine learning techniques, are trained and tested on the dataset. The classifier trained on this data is able to classify any network traffic either as normal or abnormal and provide alert outputs with details regarding the identifier of the device, type of event, risk level, and anomaly type. The alerts generated are then sent to the cross-layer correlation engine (Component 4).

5.2 Component 2: PowerShell-Based Fileless Malware Detection

Component 2 is about detection of PowerShell-based attacks using fileless malware techniques by analyzing the endpoint telemetry dataset generated via Windows Sysmon logging technology. Sysmon captures detailed information from each endpoint event such as process creations, network connection activities, registry change activities, file creation events, and PowerShell script executions.

The detection approach used utilizes a mixed AI solution consisting of Extreme Gradient Boosting (XGBoost) and Long Short-Term Memory (LSTM). XGBoost algorithm analyzes individual events by executing behavior-based classification on the extracted feature set such as the relationship between processes, the arguments being used, and other details from which malicious behavior can be identified. LSTM on the other hand analyzes sequences of events to detect multistage attacks where different stages of attacks are executed in sequence, including reconnaissance, execution, persistence, and lateral movements.

Alerts along with confidence levels and stages of the attacks are generated by the proposed model, providing insight into the stages and magnitude of detected fileless attacks. The generated alerts are used as input to the multi-layer correlation engine, which reconstructs the attack path.

5.3 Component 3: Predictive IoMT Threat Forecasting Engine

The third component examines the traffic patterns of behaviors in IoMT in order to predict the likelihood of future threats and malicious activities in the healthcare IoMT domain. This will allow proactive defense against malicious attacks, rather than reactive defense after an attack occurs.

The CICIoMT2024 dataset acts as the primary data source, where traffic flows are categorized either as a normal flow or an attacked flow. The preprocessing phase involves deleting any missing values, removing duplicates, standardizing the dataset format, and separating features from labels. A random forest classifier is applied on the extracted behavioral features of IoMT traffic flows, providing 99% accurate classification. Through feature importance analysis based on Random Forest, 20 most important behavioral attributes out of 45 features can be found.

The risk assessment process based on probabilities translates the probability predictions of the model into risk levels (Low, Medium, High) and alert statuses (SAFE, WARNING, ALERT) automatically. Models based on Long Short-Term Memory (LSTM) networks are considered as extensions of the predictive models to allow future prediction in addition to current prediction based on sequential patterns of behavior..

5.4 Component 4: Cross-Layer Lateral Movement Correlation Detection Engine and XAI Dashboard

Component 4 acts as the layer for integrating and analyzing the entire hybrid detection framework. This component correlates alerts generated by Components 1, 2, and 3 to detect attack progression within cross-layer attacks that utilize lateral movement techniques used in fileless malware attacks on IoMT health care infrastructure.

The correlation module takes the structured alert data from all detection modules as inputs and compares this data to MITRE ATT&CK TTPs applicable to health care environments. The event correlation is accomplished using a graph-based approach whereby the attack progression graph is built with IoMT devices, endpoints, and network sections as nodes and malicious interactions as edges.

The machine learning models that are applied to the correlation graph categorize the event clusters discovered into categories of being normal, suspicious, or lateral movement attacks. The XAI dashboard is designed in such a way that it employs both the SHAP and LIME explanations, which are used to explain the logic behind any detection process. The dashboard provides such details as attack progression, lateral movements, SHAP feature importance, explanation of events, risk level, and the timeline of investigations.

The whole platform is designed using a web-based platform where security analysts at the healthcare SOC can access the interface without having to install any software.

6. Sources for Test Data and Analysis

6.1 Datasets

Dataset	Used By	Purpose
CICIoMT2024	Components 1, 3	IoMT network traffic – normal and attack flows for anomaly detection and threat forecasting
Sysmon Endpoint Telemetry Logs	Component 2	Windows endpoint events for PowerShell-based fileless malware behavioral analysis
MITRE ATT&CK Knowledge Base	Component 4	TTP mapping for lateral movement correlation and attack path classification
Synthetic IoMT Lateral Movement Scenarios	Component 4	Simulated multi-stage attack sequences for cross-layer correlation engine evaluation
UNSW-NB15 / CICIDS2017 (supplementary)	Components 1, 4	Supplementary labeled network intrusion datasets for model generalization testing

6.2 Data Collection Procedures

- Network traffic datasets are accessed in their published CSV or PCAP formats from respective repositories.
- Sysmon logs are collected from controlled Windows environments configured with fileless malware simulation scripts replicating PowerShell-based attack techniques.
- Synthetic lateral movement scenarios are constructed by scripting sequential attack events across simulated IoMT network topologies.
- All datasets undergo cryptographic hashing (SHA-256) prior to processing to ensure integrity and reproducibility.

6.3 Data Analysis Methods

- Preprocessing: missing value removal, deduplication, format normalization, label encoding, and feature scaling.
- Feature engineering: extraction of behavioral, statistical, and temporal features from raw network flows and endpoint logs.
- Model training and evaluation using stratified k-fold cross-validation to ensure robust performance estimates.
- Quantitative evaluation metrics: Accuracy, Precision, Recall, F1-Score, False Positive Rate, and AUC-ROC.
- XAI evaluation: SHAP and LIME explanation consistency and analyst usability assessment through structured review.

7. Anticipated Benefits

7.1 Benefits to Users

- Healthcare security operations analysts gain a unified, multi-layer detection and investigation dashboard that reduces manual correlation effort and improves threat response time.
- Hospital IT security teams benefit from proactive threat forecasting that enables defensive action before attacks fully materialize.
- Clinical staff are indirectly protected through improved detection of attacks targeting medical imaging systems, patient monitoring devices, and hospital networks.
- Security researchers gain a validated, open architecture for hybrid AI-based IoMT security analysis that can be extended and adapted to new healthcare threat scenarios.

7.2 Contribution to the Body of Knowledge

- Provides the first integrated cross-layer hybrid AI detection framework specifically designed for fileless malware in IoMT healthcare environments.
- Demonstrates the effectiveness of combining DICOM network anomaly detection, endpoint behavioral analysis, predictive threat forecasting, and XAI-driven lateral movement correlation in a unified architecture.
- Contributes validated ML model configurations and evaluation benchmarks for IoMT security research using the CICIoMT2024 dataset.
- Establishes an XAI methodology for making AI-driven IoMT security decisions interpretable and suitable for use in regulated healthcare environments.

8. Scope and Specified Deliverables / Expected Research Outcomes

8.1 Scope

This research encompasses the design, development, and evaluation of a four-module hybrid AI detection framework targeting fileless malware threats in IoMT healthcare environments. The framework operates on collected network traffic data and endpoint telemetry logs representing realistic healthcare IoMT scenarios.

The following are within scope:

- DICOM network traffic anomaly detection using ML classifiers.
- PowerShell-based fileless malware detection using hybrid XGBoost and LSTM models applied to Sysmon telemetry.
- IoMT behavioral threat forecasting using Random Forest and LSTM-based time-series models.
- Cross-layer event correlation and lateral movement detection using graph-based analysis.
- XAI dashboard with SHAP and LIME explanations, attack graphs, and risk visualization.
- Evaluation of all modules using quantitative performance metrics.

The following are outside scope:

- Real-time deployment within live hospital network infrastructure.
- Physical IoMT device integration or hardware-level analysis.
- Automated threat response or remediation actions.
- Compliance certification or regulatory audit support.

8.2 Expected Deliverables

Deliverable	Description
DICOM Anomaly Detection Module	Trained ML classifier with alert generation for DICOM network traffic anomalies
Fileless Malware Detection Module	Hybrid XGBoost + LSTM system detecting PowerShell-based attacks from Sysmon logs
Threat Forecasting Engine	Random Forest + LSTM predictive model with risk-tiered alerting and Streamlit dashboard
Correlation Detection Engine	Graph-based cross-layer event correlator identifying lateral movement paths across IoMT layers
XAI Security Dashboard	Web-based dashboard with SHAP/LIME explanations, attack graphs, and investigation timeline
Evaluation Report	Quantitative performance analysis of all modules with benchmark comparisons
PPR, SRS, and Final Thesis	Academic documentation conforming to departmental submission requirements

9. Research Constraints

- Limited availability of real-world labeled IoMT attack datasets — mitigated by using CICIoMT2024 and generating controlled synthetic scenarios.
- Absence of live hospital network access — all experiments conducted on collected datasets and simulated network environments.
- Computational requirements for LSTM and graph-based ML model training — managed through optimized feature selection and phased training pipelines.
- Variability in Sysmon log formats across different Windows environments — addressed through normalization preprocessing pipelines.
- XAI explanation computational overhead — SHAP and LIME computations optimized for dashboard responsiveness using sampling and caching strategies.
- Ethical and data protection constraints — all datasets used are publicly available or synthetically generated; no real patient data is processed.

10. Project Plan / Schedule

Phase	Activity	Timeline	Responsible
Phase 1 – Research & Design	Literature review, problem definition, architecture design	Completed (PP1)	All Members
Phase 2 – Component 1 Development	DICOM dataset preprocessing, ML model training, alert output	Month 1–2	Student 1
Phase 3 – Component 2 Development	Sysmon log collection, XGBoost + LSTM hybrid model, alert generation	Month 1–2	Student 2
Phase 4 – Component 3 Development	CICIoMT2024 preprocessing, RF + LSTM forecasting, dashboard	Month 1–2	Student 3
Phase 5 – Component 4 Development	Correlation engine, graph analysis, XAI dashboard integration	Month 2–3	Student 4
Phase 6 – Integration & Testing	Module integration, end-to-end testing, performance evaluation	Month 3–4	All Members
Phase 7 – Documentation	SRS, compliance report, thesis writing, final PPR preparation	Month 4–5	All Members
Phase 8 – PP2 Submission	Final demonstration, report submission, viva preparation	Month 5–6	All Members

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Appendices

Appendix A – System Architecture Overview

The system follows a modular web-based architecture. Three detection components (DICOM Anomaly Detection, Fileless Malware Detection, Threat Forecasting Engine) independently process their respective data sources and generate structured alert outputs. The Cross-Layer Lateral Movement Correlation Engine aggregates these alerts, performs graph-based attack path analysis, and applies machine learning-based classification. The XAI Dashboard, implemented as a web application, presents detection results, SHAP/LIME explanations, lateral movement graphs, and risk levels to security analysts through a browser-accessible interface. All components communicate via RESTful API endpoints, and alert data is persisted in a central database for correlation and audit trail purposes.

Appendix B – Dataset Summary

Dataset	Source	Records (Approx.)	Format
CICIoMT2024	University of New Brunswick – CIC	~1.8M flows	CSV
Sysmon Endpoint Logs	Controlled Windows lab environment	Variable	XML / EVTX
MITRE ATT&CK TTPs	MITRE Corporation	600+ techniques	JSON / STIX
Synthetic Lateral Movement Logs	Research-generated scenarios	Variable	CSV / JSON

Appendix C – Module Responsibilities

Component	Module	Student
Component 1	DICOM Medical Imaging Network Anomaly Detection	[STUDENT NAME 1]
Component 2	PowerShell-Based Fileless Malware Detection	[STUDENT NAME 2]
Component 3	Predictive IoMT Threat Forecasting Engine	[STUDENT NAME 3]
Component 4	Cross-Layer Lateral Movement Correlation Detection Engine & XAI Dashboard	[STUDENT NAME 4]